CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

SILVAS HOLSTEINS

6706 ELAINE RD

TURLOCK CA 95380

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Silvas Holsteins Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
6706 Elaine RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X057-X013-X019-XXXX X057-X220-X010-XXXX X057-X220-X012-XXXX

B. OPERATORS

Silva, Adrian J

Operator name: Silva, Adrian J

Telephone no.: (209) 632-1223 Landline (209) 595-1846 Cellular

Mailing Address Number and Street	City	State	Zip Code
6706 Elaine RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

Silva, Manuel M

Operator name: Silva, Manuel M

Telephone no.: (209) 632-1223 Landline (209) 595-1846 Cellular

Mailing Address Number and Street	City	State	Zip Code
6706 Elaine RD	Turlock	CA	95380

C. OWNERS

Silva, Adrian J

Legal owner name:	Silva, Adrian J	Telephone no.:	(209) 632-1223	(209) 595-1846
			Landline	Cellular
6706 Elaine RD	Turlock		CA	95380
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

Silva, Manuel M

Legal owner name:	Silva, Manuel M	Telephone no.:	(209) 632-1223	(209) 595-1846
			Landline	Cellular
6706 Elaine RD	Turlock		CA	95380
Mailing Address Number and Street	City		State	Zip Code

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	211	0	0	0	0
Number under roof	1,233	0	0	738	155	163
Maximum number	1,233	211	0	738	155	163
Average number	1,233	211	0	738	155	163
Avg live weight (lbs)	1,400	1,500	0	900		

Predominant milk cow breed: Holstein

Average milk production: 79 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 44,638.74 *tons per reporting period*

Total nitrogen from manure: 555,345.07 *lbs per reporting period*

After ammonia losses (30% loss applied): 388,741.55 *lbs per reporting period*

Total phosphorus from manure: 91,222.40 *lbs per reporting period*

Total potassium from manure: 243,421.61 *lbs per reporting period*

Total salt from manure: 629,077.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 20,272,208 *gallons*

Total nitrogen generated: 155,625.35 *lbs*

Total phosphorus generated: 31,957.24 *lbs*

Total potassium generated: 272,281.68 *lbs*

Total salt generated: 1,034,706.81 *lbs*

	10,120,208	*gallons applied*
+	10,152,000	*gallons exported*
-	0	*gallons imported*
=	20,272,208	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
06/04/2023	Process wastewater	2,880,000.00 *gal*	1,140.00	488.00	0.00	3.10	214.00	2,670.00		10,100
06/15/2023	Process wastewater	2,520,000.00 *gal*	1,140.00	488.00	0.00	3.10	214.00	2,670.00		10,100
07/22/2023	Process wastewater	2,592,000.00 *gal*	1,210.00	413.00	0.00	1.10	266.00	1,410.00		2,200
10/15/2023	Process wastewater	2,160,000.00 *gal*	1,210.00	413.00	0.00	1.10	266.00	1,410.00		2,200

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	99,538.22	20,191.83	176,232.38	542,378.27
Total exports for all materials	99,538.22	20,191.83	176,232.38	542,378.27

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Espindula	37	37	2	manure	X057-X022-X013-XXXX
Field 1	18	18	2	process wastewater	X057-X013-X019-XXXX
Field 10	19	19	0	manure	X044-X041-X009-XXXX
Field 11	8	8	2	manure	X057-X013-X009-XXXX
Field 2	15	15	2	manure	X057-X013-X010-XXXX
Field 3	26	26	2	both	X057-X022-X012-XXXX
Field 4	27	27	2	both	X057-X022-X012-XXXX
Field 5	24	24	2	both	X057-X022-X001-XXXX X057-X022-X012-XXXX
Field 6	38	38	2	both	X057-X013-X014-XXXX
Field 9	13	13	0	manure	X044-X041-X008-XXXX
Totals for areas that were used for application	225	225	16		
Totals for areas that were not used for application					
Land application area totals	225	225	16		

B. CROPS AND HARVESTS

Espindula

Field name: Espindula

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	590.00 *ton*	Dry-weight		57.1	18,600.00	4,200.00	31,300.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.95	254.48	57.46	428.23	1,108.21

Espindula

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	1,218.40 *ton*	Dry-weight		64.7	11,800.00	2,700.00	14,400.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.93	274.33	62.77	334.78	1,208.92

Field 1

Field name: Field 1

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	296.90 *ton*	Dry-weight		70.0	24,500.00	4,200.00	40,900.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.49	242.47	41.57	404.77	1,118.32

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	494.40 *ton*	Dry-weight		67.6	12,100.00	3,100.00	16,600.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.47	215.36	55.18	295.45	1,121.30

Field 10

Field name: Field 10

01/23/2021: Almond, in shell

Crop: Almond, in shell Acres planted: 19 Plant date: 01/23/2021

No harvests entered for this crop.

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.50	195.00	30.00	150.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Field 11

Field name: Field 11

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	97.00 *ton*	Dry-weight		58.9	24,500.00	4,200.00	39,200.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.13	244.19	41.86	390.70	1,046.51

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	232.50 *ton*	Dry-weight		64.7	11,800.00	2,700.00	14,400.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.06	242.11	55.40	295.46	1,066.94

Field 2

Field 2

Field name: Field 2

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	247.40 *ton*	Dry-weight		70.0	24,500.00	4,200.00	39,200.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.49	242.45	41.56	387.92	1,039.08

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	414.20 *ton*	Dry-weight		64.7	11,800.00	2,700.00	14,400.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.61	230.04	52.64	280.73	1,013.74

Field 3

Field name: Field 3

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	494.00 *ton*	Dry-weight		62.9	20,700.00	4,100.00	39,100.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.00	291.83	57.80	551.23	1,452.09

Field 3

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	704.50 *ton*	Dry-weight		66.9	11,400.00	2,400.00	13,200.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.10	204.49	43.05	236.78	914.82

Field 4

Field name: Field 4

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	469.00 *ton*	Dry-weight		62.3	19,600.00	4,300.00	34,300.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.37	256.71	56.32	449.24	1,152.56

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	745.60 *ton*	Dry-weight		64.3	12,500.00	2,900.00	13,600.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.61	246.46	57.18	268.15	1,084.43

Field 5

Field name: Field 5

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	422.00 *ton*	Dry-weight		60.3	19,600.00	4,300.00	34,300.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.58	273.64	60.03	478.87	1,228.58

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	665.50 *ton*	Dry-weight		64.3	12,500.00	2,900.00	13,600.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.73	247.48	57.42	269.26	1,088.92

Field 6

Field name: Field 6

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	738.00 *ton*	Dry-weight		64.9	20,600.00	4,300.00	38,700.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.42	280.85	58.62	527.62	1,390.63

Field 6

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	809.30 *ton*	Dry-weight		67.6	12,100.00	3,100.00	16,600.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.30	166.99	42.78	229.09	869.44

Field 9

Field name: Field 9

01/23/2021: Almond, in shell

Crop: Almond, in shell Acres planted: 13 Plant date: 01/23/2021

No harvests entered for this crop.

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.50	195.00	30.00	150.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Espindula - 10/03/2022: Oats, silage-soft dough

Field name: Espindula

Crop: Oats, silage-soft dough — Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	330.83	45.95	140.14	0.00	269.00 *ton*
Application event totals		330.83	45.95	140.14	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	13.73	0.00	0.00	365.43	9,508,491.00 *gal*
Application event totals		13.73	0.00	0.00	365.43	

Espindula - 05/13/2023: Corn, silage

Field name: Espindula

Crop: Corn, silage — Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	284.58	116.48	577.10	0.00	537.00 *ton*
Application event totals		284.58	116.48	577.10	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.45	0.00	0.00	304.78	7,930,288.00 *gal*
Application event totals		11.45	0.00	0.00	304.78	

Espindula - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.74	0.00	0.00	285.96	7,440,753.00 *gal*
Application event totals		10.74	0.00	0.00	285.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.61	0.00	0.00	255.86	6,657,518.00 *gal*
Application event totals		9.61	0.00	0.00	255.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.50	0.00	0.00	279.64	7,276,272.00 *gal*
Application event totals		10.50	0.00	0.00	279.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.03	0.00	0.00	267.15	6,951,219.00 *gal*
Application event totals		10.03	0.00	0.00	267.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.02	0.00	0.00	266.70	6,939,486.00 *gal*
Application event totals		10.02	0.00	0.00	266.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.63	0.00	0.00	256.31	6,669,250.00 *gal*
Application event totals		9.63	0.00	0.00	256.31	

Espindula - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.03	0.00	0.00	267.15	6,951,219.00 *gal*
Application event totals		10.03	0.00	0.00	267.15	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.00	0.00	0.00	239.61	6,234,565.00 *gal*
Application event totals		9.00	0.00	0.00	239.61	

Field 1 - 10/03/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.90	12.75	149.24	937.31	300,857.00 *gal*
Canal	Surface water	9.82	0.00	0.00	261.43	3,309,254.00 *gal*
Application event totals		114.72	12.75	149.24	1,198.74	

03/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.30	17.91	120.54	526.11	243,000.00 *gal*
Application event totals		104.30	17.91	120.54	526.11	

Field 1 - 10/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.06	13.09	186.69	815.43	231,429.00 *gal*
Canal	Surface water	8.58	0.00	0.00	228.48	2,892,167.00 *gal*
Application event totals		112.64	13.09	186.69	1,043.91	

Field 1 - 05/13/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.83	5.89	84.01	366.94	104,143.00 *gal*
Canal	Surface water	9.75	0.00	0.00	259.59	3,285,952.00 *gal*
Application event totals		56.58	5.89	84.01	626.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.26	0.00	0.00	220.06	2,785,532.00 *gal*
Application event totals		8.26	0.00	0.00	220.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.83	5.89	84.01	366.94	104,143.00 *gal*
Canal	Surface water	11.50	0.00	0.00	306.25	3,876,613.00 *gal*
Application event totals		58.33	5.89	84.01	673.19	

Field 1 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.32	0.00	0.00	274.76	3,478,005.00 *gal*
Application event totals		10.32	0.00	0.00	274.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.47	12.84	68.08	106.22	104,143.00 *gal*
Canal	Surface water	8.05	0.00	0.00	214.29	2,712,531.00 *gal*
Application event totals		66.52	12.84	68.08	320.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.62	0.00	0.00	229.49	2,905,007.00 *gal*
Application event totals		8.62	0.00	0.00	229.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.20	0.00	0.00	218.21	2,762,231.00 *gal*
Application event totals		8.20	0.00	0.00	218.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.47	12.84	68.08	106.22	104,143.00 *gal*
Canal	Surface water	9.17	0.00	0.00	244.03	3,089,043.00 *gal*
Application event totals		67.64	12.84	68.08	350.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.85	0.00	0.00	262.27	3,319,878.00 *gal*
Application event totals		9.85	0.00	0.00	262.27	

Field 10 - 01/23/2021: Almond, in shell

Field name: Field 10

Crop: Almond, in shell

Plant date: 01/23/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	29.03	387,854.00 *gal*
Application event totals		1.09	0.00	0.00	29.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.73	0.00	0.00	19.35	258,569.00 *gal*
Application event totals		0.73	0.00	0.00	19.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.54	0.00	0.00	67.73	904,959.00 *gal*
Application event totals		2.54	0.00	0.00	67.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.70	0.00	0.00	45.16	603,360.00 *gal*
Application event totals		1.70	0.00	0.00	45.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.70	0.00	0.00	45.15	603,306.00 *gal*
Application event totals		1.70	0.00	0.00	45.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.70	0.00	0.00	45.15	603,306.00 *gal*
Application event totals		1.70	0.00	0.00	45.15	

Field 10 - 01/23/2021: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.54	0.00	0.00	67.73	904,959.00 *gal*
Application event totals		2.54	0.00	0.00	67.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.70	0.00	0.00	45.15	603,306.00 *gal*
Application event totals		1.70	0.00	0.00	45.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.70	0.00	0.00	45.15	603,306.00 *gal*
Application event totals		1.70	0.00	0.00	45.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.06	0.00	0.00	54.83	732,591.00 *gal*
Application event totals		2.06	0.00	0.00	54.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.00	0.00	0.00	292.96	3,914,385.00 *gal*
Application event totals		11.00	0.00	0.00	292.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.94	0.00	0.00	51.61	689,539.00 *gal*
Application event totals		1.94	0.00	0.00	51.61	

Field 10 - 01/23/2021: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.94	0.00	0.00	51.61	689,539.00 *gal*
Application event totals		1.94	0.00	0.00	51.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.94	0.00	0.00	51.61	689,539.00 *gal*
Application event totals		1.94	0.00	0.00	51.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.94	0.00	0.00	51.61	689,539.00 *gal*
Application event totals		1.94	0.00	0.00	51.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	151.70	62.09	307.64	0.00	147.00 *ton*
Application event totals		151.70	62.09	307.64	0.00	

Field 11 - 10/03/2022: Oats, silage-soft dough

Field name: Field 11

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	312.84	43.45	132.52	0.00	55.00 *ton*
Application event totals		312.84	43.45	132.52	0.00	

Field 11 - 10/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.67	0.00	0.00	417.08	2,346,480.00 *gal*
Application event totals		15.67	0.00	0.00	417.08	

Field 11 - 05/13/2023: Corn, silage

Field name: Field 11

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	262.26	107.34	531.83	0.00	107.00 *ton*
Application event totals		262.26	107.34	531.83	0.00	

05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.81	0.00	0.00	234.61	1,319,895.00 *gal*
Application event totals		8.81	0.00	0.00	234.61	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.83	0.00	0.00	208.54	1,173,240.00 *gal*
Application event totals		7.83	0.00	0.00	208.54	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.87	0.00	0.00	156.41	879,930.00 *gal*
Application event totals		5.87	0.00	0.00	156.41	

Field 11 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.08	0.00	0.00	2.09	11,732.00 *gal*
Application event totals		0.08	0.00	0.00	2.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.87	0.00	0.00	156.41	879,930.00 *gal*
Application event totals		5.87	0.00	0.00	156.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.83	0.00	0.00	208.54	1,173,240.00 *gal*
Application event totals		7.83	0.00	0.00	208.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.83	0.00	0.00	208.54	1,173,240.00 *gal*
Application event totals		7.83	0.00	0.00	208.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	13.71	0.00	0.00	364.95	2,053,170.00 *gal*
Application event totals		13.71	0.00	0.00	364.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.83	0.00	0.00	208.54	1,173,240.00 *gal*
Application event totals		7.83	0.00	0.00	208.54	

Field 2 - 10/03/2022: Oats, silage-soft dough

Field 2 - 10/03/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	315.49	43.82	133.65	0.00	104.00 *ton*
Application event totals		315.49	43.82	133.65	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.09	0.00	0.00	215.49	2,273,117.00 *gal*
Application event totals		8.09	0.00	0.00	215.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.09	0.00	0.00	215.49	2,273,117.00 *gal*
Application event totals		8.09	0.00	0.00	215.49	

Field 2 - 05/13/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	241.83	98.98	490.41	0.00	185.00 *ton*
Application event totals		241.83	98.98	490.41	0.00	

Field 2 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.61	0.00	0.00	255.89	2,699,329.00 *gal*
Application event totals		9.61	0.00	0.00	255.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.13	0.00	0.00	296.30	3,125,541.00 *gal*
Application event totals		11.13	0.00	0.00	296.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.08	0.00	0.00	188.55	1,988,961.00 *gal*
Application event totals		7.08	0.00	0.00	188.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.04	0.00	0.00	187.48	1,977,607.00 *gal*
Application event totals		7.04	0.00	0.00	187.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.09	0.00	0.00	215.49	2,273,117.00 *gal*
Application event totals		8.09	0.00	0.00	215.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.09	0.00	0.00	215.49	2,273,117.00 *gal*
Application event totals		8.09	0.00	0.00	215.49	

Field 2 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.09	0.00	0.00	215.50	2,273,244.00 *gal*
Application event totals		8.09	0.00	0.00	215.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.13	0.00	0.00	296.30	3,125,541.00 *gal*
Application event totals		11.13	0.00	0.00	296.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.02	0.00	0.00	53.87	568,269.00 *gal*
Application event totals		2.02	0.00	0.00	53.87	

Field 3 - 11/10/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	204.77	28.44	86.74	0.00	117.00 *ton*
Application event totals		204.77	28.44	86.74	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.04	21.07	159.18	905.16	486,234.00 *gal*
Application event totals		85.04	21.07	159.18	905.16	

Field 3 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.04	21.07	159.18	905.16	486,234.00 *gal*
Canal	Surface water	16.07	0.00	0.00	427.87	7,823,215.00 *gal*
Application event totals		101.11	21.07	159.18	1,333.03	

Field 3 - 05/13/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.15	13.17	99.49	565.73	303,896.00 *gal*
Canal	Surface water	12.10	0.00	0.00	322.10	5,889,321.00 *gal*
Application event totals		65.25	13.17	99.49	887.82	

06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.78	0.00	0.00	153.92	2,814,247.00 *gal*
Application event totals		5.78	0.00	0.00	153.92	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.15	13.17	99.49	565.73	303,896.00 *gal*
Canal	Surface water	3.60	0.00	0.00	95.76	1,750,858.00 *gal*
Application event totals		56.75	13.17	99.49	661.48	

Field 3 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.81	0.00	0.00	181.27	3,314,398.00 *gal*
Application event totals		6.81	0.00	0.00	181.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.27	13.69	103.47	588.36	316,052.00 *gal*
Canal	Surface water	6.06	0.00	0.00	161.28	2,948,820.00 *gal*
Application event totals		61.33	13.69	103.47	749.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.25	0.00	0.00	219.63	4,015,720.00 *gal*
Application event totals		8.25	0.00	0.00	219.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.05	0.00	0.00	134.40	2,457,352.00 *gal*
Application event totals		5.05	0.00	0.00	134.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.53	14.75	111.43	633.61	340,364.00 *gal*
Canal	Surface water	5.30	0.00	0.00	141.12	2,580,216.00 *gal*
Application event totals		64.83	14.75	111.43	774.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.05	0.00	0.00	134.40	2,457,352.00 *gal*
Application event totals		5.05	0.00	0.00	134.40	

Field 4 - 11/10/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	165.16	22.94	69.96	0.00	98.00 *ton*
Application event totals		165.16	22.94	69.96	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.04	21.07	159.18	905.16	504,935.00 *gal*
Application event totals		85.04	21.07	159.18	905.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.04	21.07	159.18	905.16	504,935.00 *gal*
Canal	Surface water	16.07	0.00	0.00	427.87	8,124,108.00 *gal*
Application event totals		101.11	21.07	159.18	1,333.03	

Field 4 - 05/13/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.78	15.80	119.39	678.87	378,701.00 *gal*
Canal	Surface water	12.10	0.00	0.00	322.10	6,115,833.00 *gal*
Application event totals		75.88	15.80	119.39	1,000.97	

Field 4 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.78	0.00	0.00	153.92	2,922,488.00 *gal*
Application event totals		5.78	0.00	0.00	153.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.78	15.80	119.39	678.87	378,701.00 *gal*
Canal	Surface water	3.60	0.00	0.00	95.76	1,818,198.00 *gal*
Application event totals		67.38	15.80	119.39	774.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.81	0.00	0.00	181.27	3,441,875.00 *gal*
Application event totals		6.81	0.00	0.00	181.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.53	18.96	143.27	814.65	454,442.00 *gal*
Canal	Surface water	6.06	0.00	0.00	161.28	3,062,236.00 *gal*
Application event totals		82.59	18.96	143.27	975.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.25	0.00	0.00	219.63	4,170,171.00 *gal*
Application event totals		8.25	0.00	0.00	219.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.05	0.00	0.00	134.40	2,551,866.00 *gal*
Application event totals		5.05	0.00	0.00	134.40	

Field 4 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.53	18.96	143.27	814.65	454,442.00 *gal*
Canal	Surface water	5.30	0.00	0.00	141.12	2,679,455.00 *gal*
Application event totals		81.83	18.96	143.27	955.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.05	0.00	0.00	134.40	2,551,866.00 *gal*
Application event totals		5.05	0.00	0.00	134.40	

Field 5 - 11/10/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	182.02	25.28	77.10	0.00	96.00 *ton*
Application event totals		182.02	25.28	77.10	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.04	21.07	159.18	905.16	448,831.00 *gal*
Application event totals		85.04	21.07	159.18	905.16	

Field 5 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.04	21.07	159.18	905.16	448,831.00 *gal*
Canal	Surface water	16.07	0.00	0.00	427.87	7,221,429.00 *gal*
Application event totals		101.11	21.07	159.18	1,333.03	

Field 5 - 05/13/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.78	15.80	119.39	678.87	336,623.00 *gal*
Canal	Surface water	12.10	0.00	0.00	322.10	5,436,296.00 *gal*
Application event totals		75.88	15.80	119.39	1,000.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.78	0.00	0.00	153.92	2,597,767.00 *gal*
Application event totals		5.78	0.00	0.00	153.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.78	15.80	119.39	678.87	336,623.00 *gal*
Canal	Surface water	3.60	0.00	0.00	95.76	1,616,176.00 *gal*
Application event totals		67.38	15.80	119.39	774.63	

Field 5 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.81	0.00	0.00	181.27	3,059,444.00 *gal*
Application event totals		6.81	0.00	0.00	181.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.53	18.96	143.27	814.65	403,948.00 *gal*
Canal	Surface water	6.06	0.00	0.00	161.28	2,721,988.00 *gal*
Application event totals		82.59	18.96	143.27	975.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.25	0.00	0.00	219.63	3,706,819.00 *gal*
Application event totals		8.25	0.00	0.00	219.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.05	0.00	0.00	134.40	2,268,325.00 *gal*
Application event totals		5.05	0.00	0.00	134.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.53	18.96	143.27	814.65	403,948.00 *gal*
Canal	Surface water	5.30	0.00	0.00	141.12	2,381,738.00 *gal*
Application event totals		81.83	18.96	143.27	955.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.05	0.00	0.00	134.40	2,268,325.00 *gal*
Application event totals		5.05	0.00	0.00	134.40	

Field 6 - 11/10/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	164.05	22.79	69.50	0.00	137.00 *ton*
Application event totals		164.05	22.79	69.50	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.30	17.91	120.54	526.11	513,000.00 *gal*
Application event totals		104.30	17.91	120.54	526.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.06	13.09	186.69	815.42	488,571.00 *gal*
Canal	Surface water	11.68	0.00	0.00	310.94	8,309,225.00 *gal*
Application event totals		115.74	13.09	186.69	1,126.36	

Field 6 - 05/13/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.83	5.89	84.01	366.94	219,857.00 *gal*
Canal	Surface water	10.01	0.00	0.00	266.52	7,122,173.00 *gal*
Application event totals		56.84	5.89	84.01	633.46	

Field 6 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.77	0.00	0.00	47.09	1,258,274.00 *gal*
Application event totals		1.77	0.00	0.00	47.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.34	0.00	0.00	302.05	8,071,787.00 *gal*
Application event totals		11.34	0.00	0.00	302.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.63	5.24	74.68	326.17	195,429.00 *gal*
Canal	Surface water	6.61	0.00	0.00	176.12	4,706,596.00 *gal*
Application event totals		48.24	5.24	74.68	502.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.01	0.00	0.00	266.52	7,122,173.00 *gal*
Application event totals		10.01	0.00	0.00	266.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.26	0.00	0.00	299.83	8,012,473.00 *gal*
Application event totals		11.26	0.00	0.00	299.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.36	0.00	0.00	249.19	6,659,252.00 *gal*
Application event totals		9.36	0.00	0.00	249.19	

Field 6 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	4.31	0.00	0.00	114.82	3,068,485.00 *gal*
Application event totals		4.31	0.00	0.00	114.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.47	12.84	68.08	106.22	219,857.00 *gal*
Canal	Surface water	5.01	0.00	0.00	133.26	3,561,109.00 *gal*
Application event totals		63.48	12.84	68.08	239.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.46	0.00	0.00	172.13	4,599,740.00 *gal*
Application event totals		6.46	0.00	0.00	172.13	

Field 9 - 01/23/2021: Almond, in shell

Field name: Field 9

Crop: Almond, in shell

Plant date: 01/23/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.59	0.00	0.00	42.42	387,854.00 *gal*
Application event totals		1.59	0.00	0.00	42.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.12	0.00	0.00	56.57	517,138.00 *gal*
Application event totals		2.12	0.00	0.00	56.57	

Field 9 - 01/23/2021: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.48	0.00	0.00	65.99	603,306.00 *gal*
Application event totals		2.48	0.00	0.00	65.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.48	0.00	0.00	65.99	603,306.00 *gal*
Application event totals		2.48	0.00	0.00	65.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.48	0.00	0.00	65.99	603,306.00 *gal*
Application event totals		2.48	0.00	0.00	65.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.97	0.00	0.00	212.12	1,939,235.00 *gal*
Application event totals		7.97	0.00	0.00	212.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	3.72	0.00	0.00	98.99	904,959.00 *gal*
Application event totals		3.72	0.00	0.00	98.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.48	0.00	0.00	65.99	603,306.00 *gal*
Application event totals		2.48	0.00	0.00	65.99	

Field 9 - 01/23/2021: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.48	0.00	0.00	65.99	603,306.00 *gal*
Application event totals		2.48	0.00	0.00	65.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	3.01	0.00	0.00	80.13	732,591.00 *gal*
Application event totals		3.01	0.00	0.00	80.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.49	0.00	0.00	412.54	3,771,478.00 *gal*
Application event totals		15.49	0.00	0.00	412.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.83	0.00	0.00	75.42	689,539.00 *gal*
Application event totals		2.83	0.00	0.00	75.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.83	0.00	0.00	75.42	689,539.00 *gal*
Application event totals		2.83	0.00	0.00	75.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.83	0.00	0.00	75.42	689,539.00 *gal*
Application event totals		2.83	0.00	0.00	75.42	

Field 9 - 01/23/2021: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.83	0.00	0.00	75.42	689,539.00 *gal*
Application event totals		2.83	0.00	0.00	75.42	

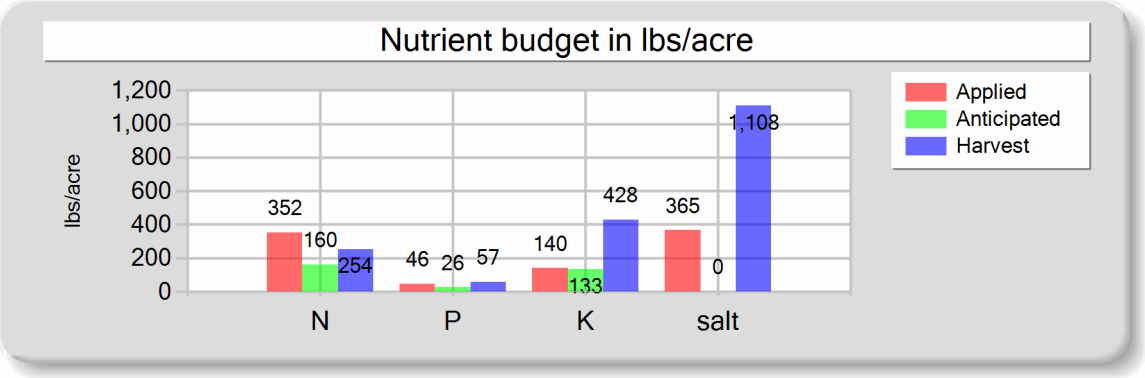
10/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	150.83	61.74	305.87	0.00	100.00 *ton*
Application event totals		150.83	61.74	305.87	0.00	

B. NUTRIENT BUDGET

Espindula - 10/03/2022: Oats, silage-soft dough

Field name: Espindula | Crop: Oats, silage-soft dough | Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	330.83	45.95	140.14	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.73	0.00	0.00	365.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	351.55	45.95	140.14	365.43
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	254.48	57.46	428.23	1,108.21
Nutrient balance	97.07	-11.51	-288.09	-742.78
Applied to removed ratio	1.38	0.80	0.33	0.33

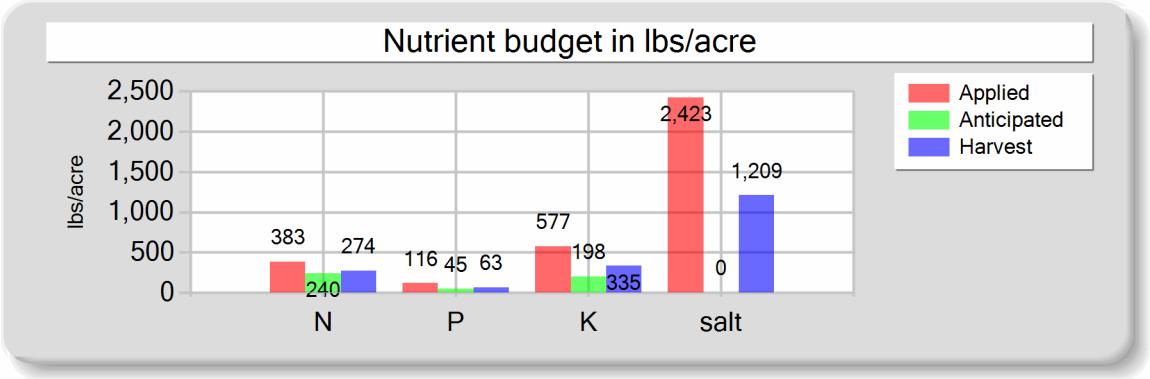
Fresh water applied	
9,508,491.00	*gallons*
350.17	*acre-inches*
9.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Espindula - 05/13/2023: Corn, silage

Field name: Espindula Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	284.58	116.48	577.10	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	91.01	0.00	0.00	2,423.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	382.59	116.48	577.10	2,423.17
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	274.33	62.77	334.78	1,208.92
Nutrient balance	108.26	53.71	242.33	1,214.25
Applied to removed ratio	1.39	1.86	1.72	2.00

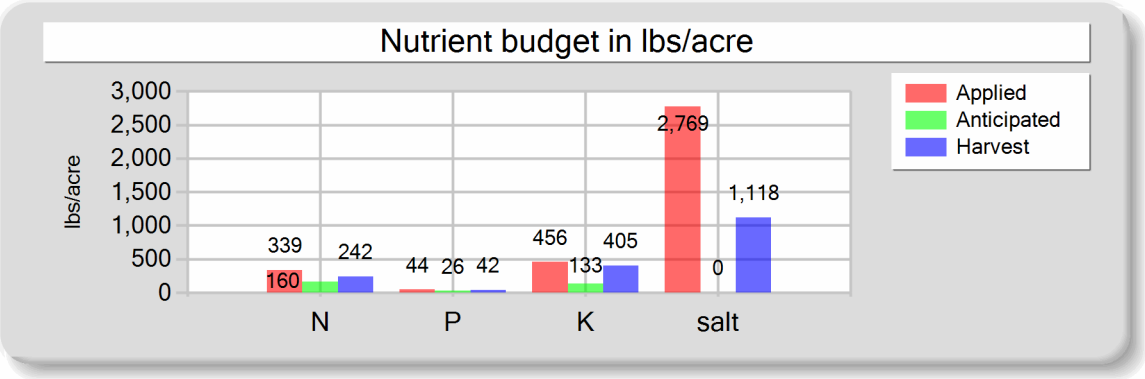
Fresh water applied	
63,050,570.00	*gallons*
2,321.94	*acre-inches*
62.76	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 10/03/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	313.27	43.75	456.48	2,278.85
Fresh water	18.40	0.00	0.00	489.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	338.67	43.75	456.48	2,768.76
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	242.47	41.57	404.77	1,118.32
Nutrient balance	96.20	2.18	51.70	1,650.43
Applied to removed ratio	1.40	1.05	1.13	2.48

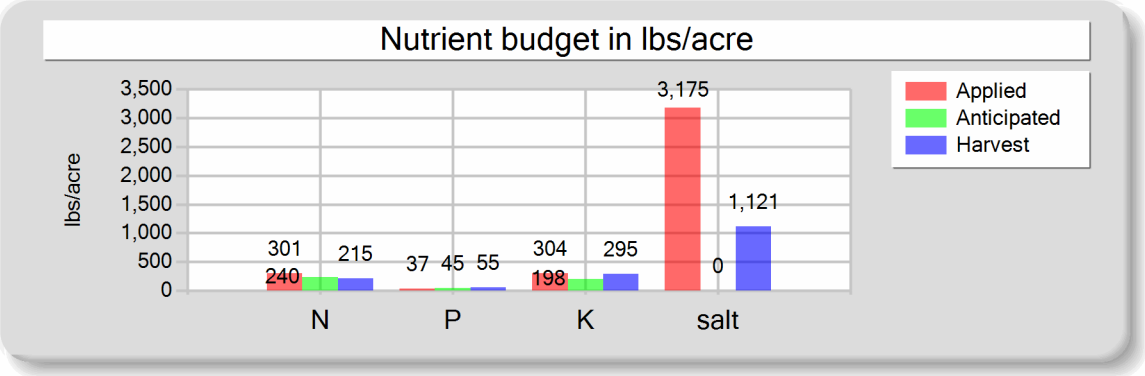
Fresh water applied	
6,201,421.00	*gallons*
228.38	*acre-inches*
12.69	*inches/acre*

Process wastewater applied	
775,286.00	*gallons*
28.55	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/13/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.61	37.47	304.18	946.32
Fresh water	83.72	0.00	0.00	2,228.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.32	37.47	304.18	3,175.27
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	215.36	55.18	295.45	1,121.30
Nutrient balance	85.96	-17.71	8.72	2,053.97
Applied to removed ratio	1.40	0.68	1.03	2.83

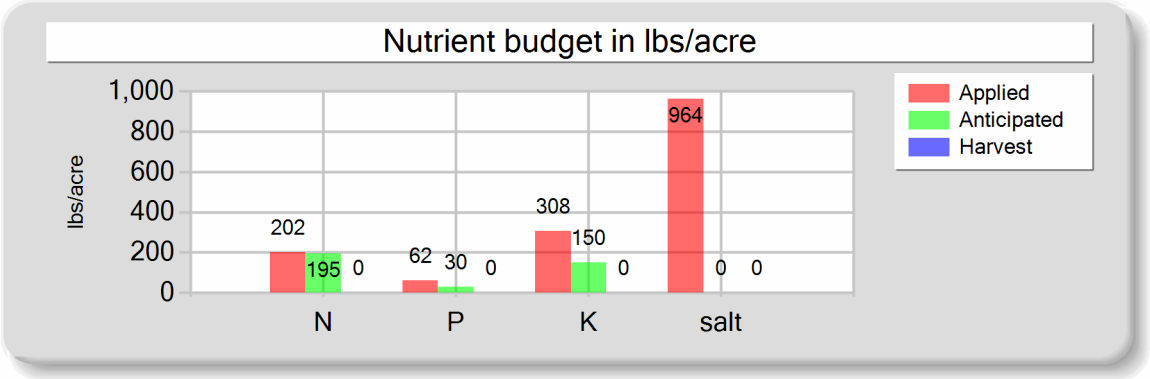
Fresh water applied	
28,214,792.00	*gallons*
1,039.05	*acre-inches*
57.73	*inches/acre*

Process wastewater applied	
416,572.00	*gallons*
15.34	*acre-inches*
0.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 10 - 01/23/2021: Almond, in shell

Field name: Field 10 Crop: Almond, in shell Plant date: 01/23/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	151.70	62.09	307.64	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	36.20	0.00	0.00	963.81
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	201.90	62.09	307.64	963.81
Anticipated crop nutrient removal	195.00	30.00	150.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	201.90	62.09	307.64	963.81
Applied to removed ratio	0.00	0.00	0.00	0.00

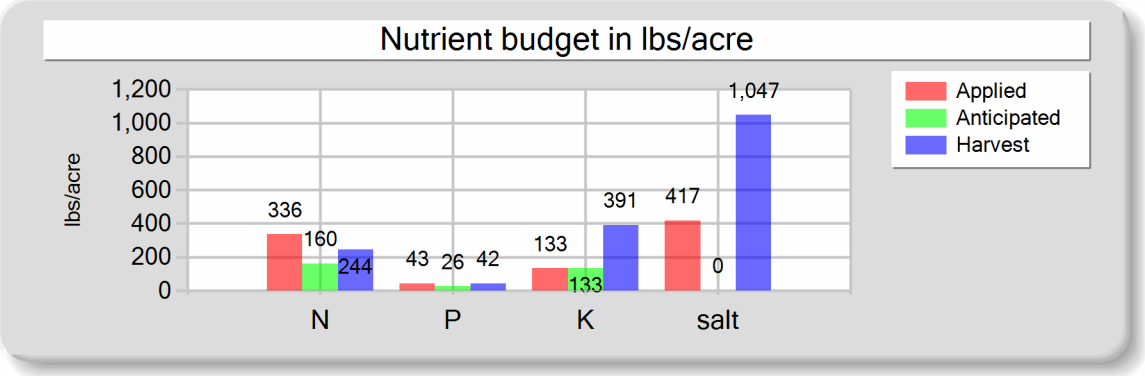
Fresh water applied	
12,878,057.00	*gallons*
474.26	*acre-inches*
24.96	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 11 - 10/03/2022: Oats, silage-soft dough

Field name: Field 11 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	312.84	43.45	132.52	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.67	0.00	0.00	417.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.51	43.45	132.52	417.08
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	244.19	41.86	390.70	1,046.51
Nutrient balance	91.32	1.59	-258.17	-629.43
Applied to removed ratio	1.37	1.04	0.34	0.40

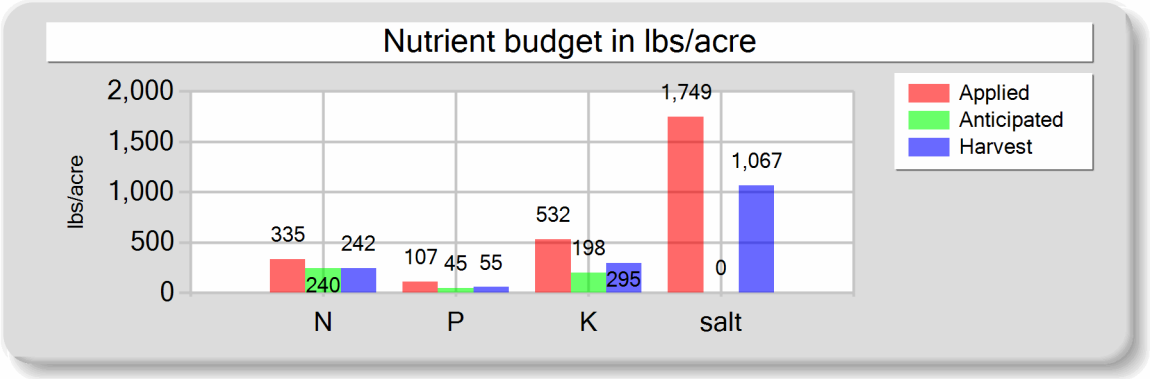
Fresh water applied	
2,346,480.00	*gallons*
86.41	*acre-inches*
10.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 11 - 05/13/2023: Corn, silage

Field name: Field 11 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	262.26	107.34	531.83	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	65.68	0.00	0.00	1,748.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	334.93	107.34	531.83	1,748.62
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	242.11	55.40	295.46	1,066.94
Nutrient balance	92.82	51.94	236.37	681.68
Applied to removed ratio	1.38	1.94	1.80	1.64

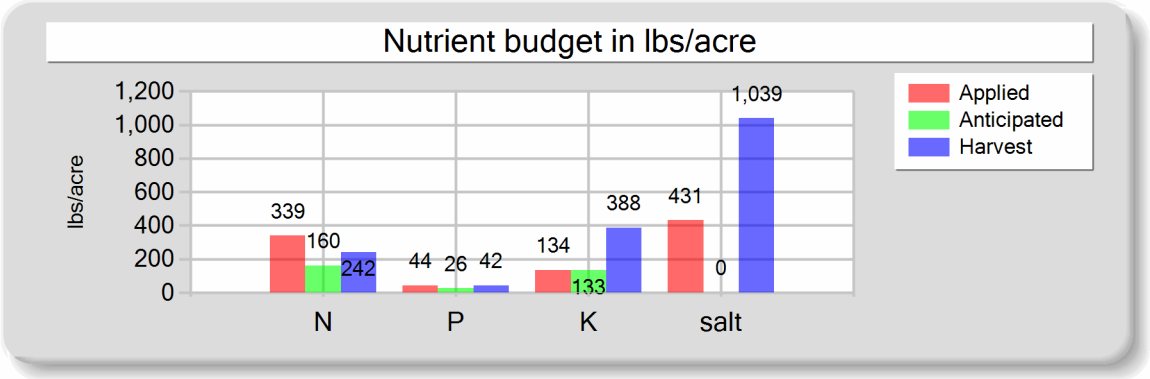
Fresh water applied
9,837,617.00 *gallons*
362.29 *acre-inches*
45.29 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 2 - 10/03/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	315.49	43.82	133.65	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	16.19	0.00	0.00	430.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	338.68	43.82	133.65	430.98
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	242.45	41.56	387.92	1,039.08
Nutrient balance	96.23	2.26	-254.28	-608.10
Applied to removed ratio	1.40	1.05	0.34	0.41

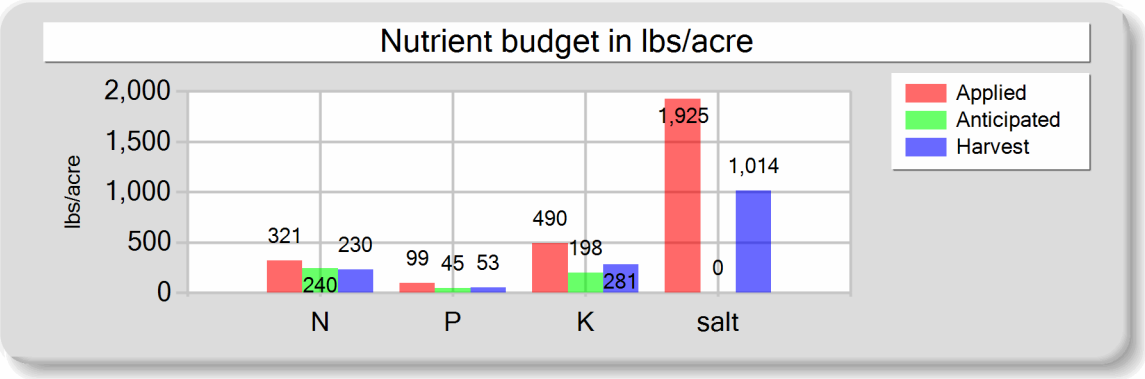
Fresh water applied	
4,546,234.00	*gallons*
167.42	*acre-inches*
11.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/13/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	241.83	98.98	490.41	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	72.30	0.00	0.00	1,924.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	321.13	98.98	490.41	1,924.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	230.04	52.64	280.73	1,013.74
Nutrient balance	91.09	46.35	209.68	911.13
Applied to removed ratio	1.40	1.88	1.75	1.90

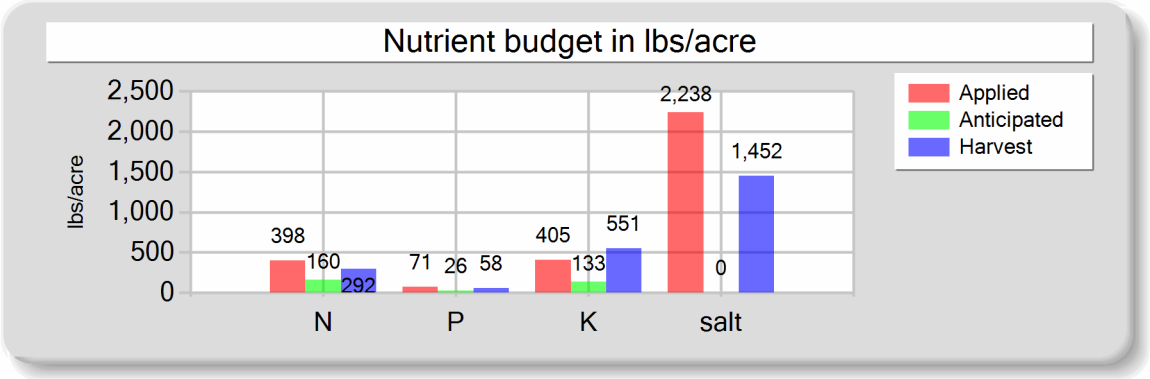
Fresh water applied	
20,304,726.00	*gallons*
747.75	*acre-inches*
49.85	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/10/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	204.77	28.44	86.74	0.00
Process wastewater	170.08	42.14	318.37	1,810.32
Fresh water	16.07	0.00	0.00	427.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	397.91	70.58	405.11	2,238.19
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	291.83	57.80	551.23	1,452.09
Nutrient balance	106.09	12.78	-146.12	786.10
Applied to removed ratio	1.36	1.22	0.73	1.54

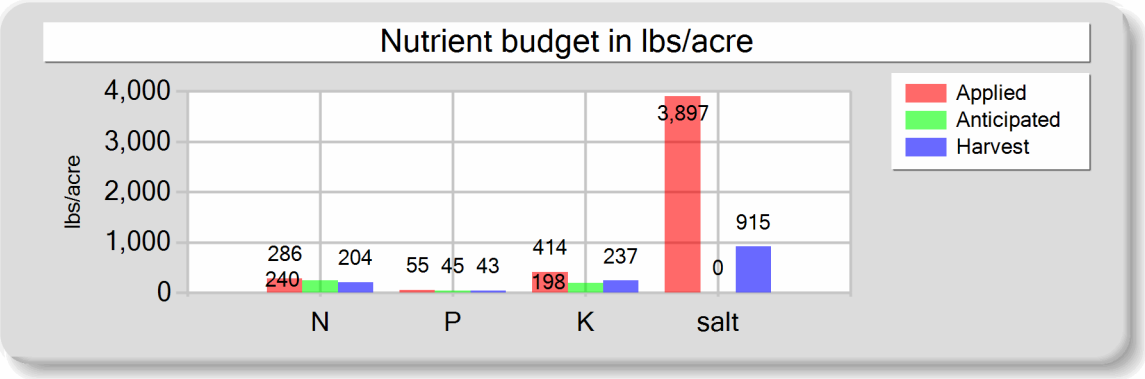
Fresh water applied	
7,823,215.00	*gallons*
288.10	*acre-inches*
11.08	*inches/acre*

Process wastewater applied	
972,468.00	*gallons*
35.81	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/13/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	221.10	54.78	413.88	2,353.42
Fresh water	57.99	0.00	0.00	1,543.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.09	54.78	413.88	3,897.28
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	204.49	43.05	236.78	914.82
Nutrient balance	81.60	11.73	177.10	2,982.46
Applied to removed ratio	1.40	1.27	1.75	4.26

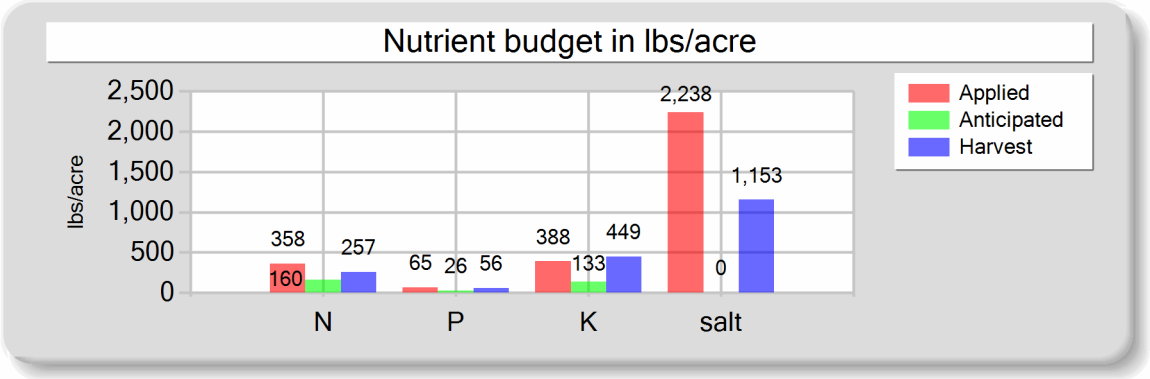
Fresh water applied	
28,228,284.00	*gallons*
1,039.55	*acre-inches*
39.98	*inches/acre*

Process wastewater applied	
1,264,208.00	*gallons*
46.56	*acre-inches*
1.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/10/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	165.16	22.94	69.96	0.00
Process wastewater	170.08	42.14	318.37	1,810.32
Fresh water	16.07	0.00	0.00	427.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	358.31	65.08	388.33	2,238.19
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	256.71	56.32	449.24	1,152.56
Nutrient balance	101.60	8.76	-60.90	1,085.63
Applied to removed ratio	1.40	1.16	0.86	1.94

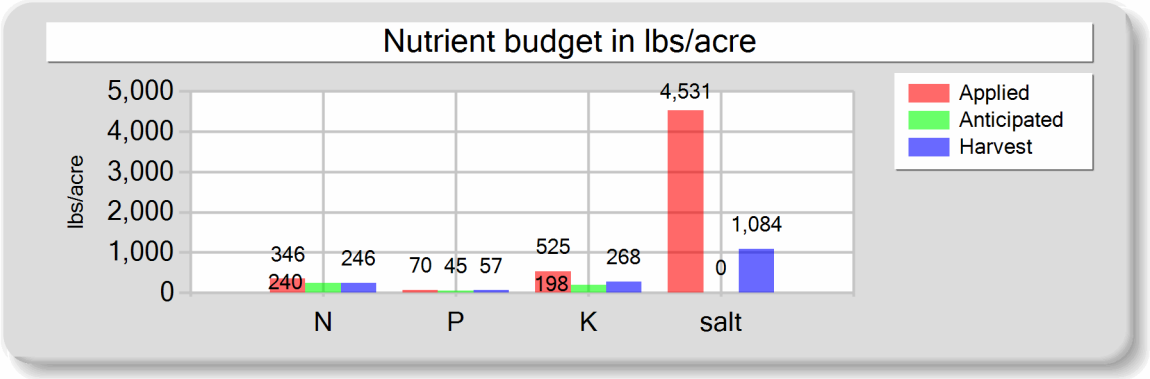
Fresh water applied	
8,124,108.00	*gallons*
299.18	*acre-inches*
11.08	*inches/acre*

Process wastewater applied	
1,009,870.00	*gallons*
37.19	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 05/13/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.63	69.53	525.31	2,987.03
Fresh water	57.99	0.00	0.00	1,543.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	345.61	69.53	525.31	4,530.89
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	246.46	57.18	268.15	1,084.43
Nutrient balance	99.15	12.35	257.16	3,446.46
Applied to removed ratio	1.40	1.22	1.96	4.18

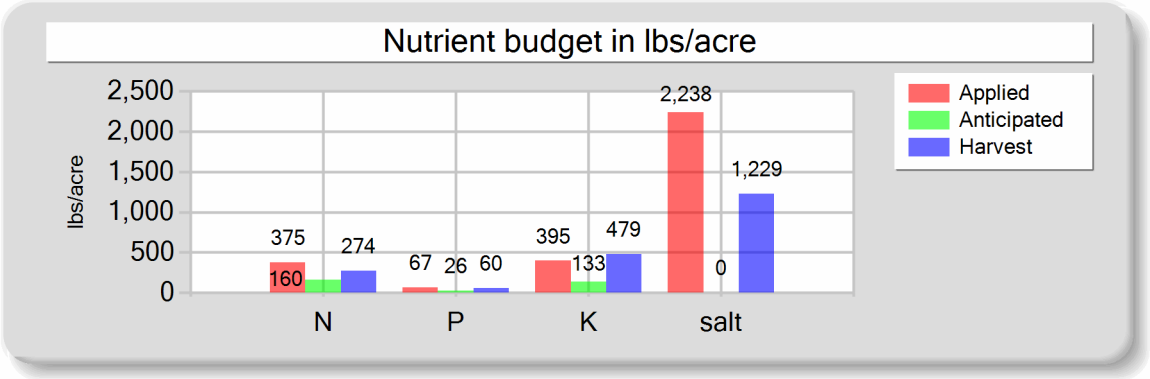
Fresh water applied	
29,313,988.00	*gallons*
1,079.53	*acre-inches*
39.98	*inches/acre*

Process wastewater applied	
1,666,286.00	*gallons*
61.36	*acre-inches*
2.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/10/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	182.02	25.28	77.10	0.00
Process wastewater	170.08	42.14	318.37	1,810.32
Fresh water	16.07	0.00	0.00	427.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	375.16	67.42	395.47	2,238.19
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	273.64	60.03	478.87	1,228.58
Nutrient balance	101.52	7.38	-83.40	1,009.61
Applied to removed ratio	1.37	1.12	0.83	1.82

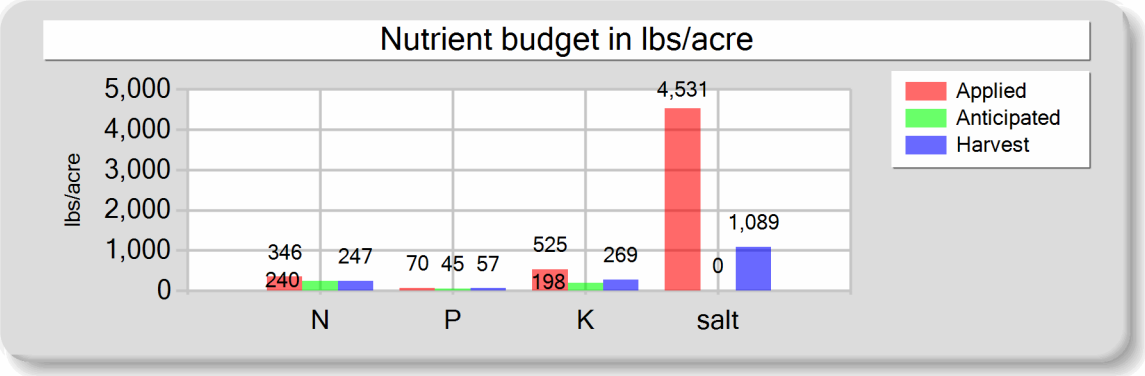
Fresh water applied	
7,221,429.00	*gallons*
265.94	*acre-inches*
11.08	*inches/acre*

Process wastewater applied	
897,662.00	*gallons*
33.06	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 05/13/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.63	69.53	525.31	2,987.03
Fresh water	57.99	0.00	0.00	1,543.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	345.61	69.53	525.31	4,530.89
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	247.48	57.42	269.26	1,088.92
Nutrient balance	98.13	12.11	256.04	3,441.96
Applied to removed ratio	1.40	1.21	1.95	4.16

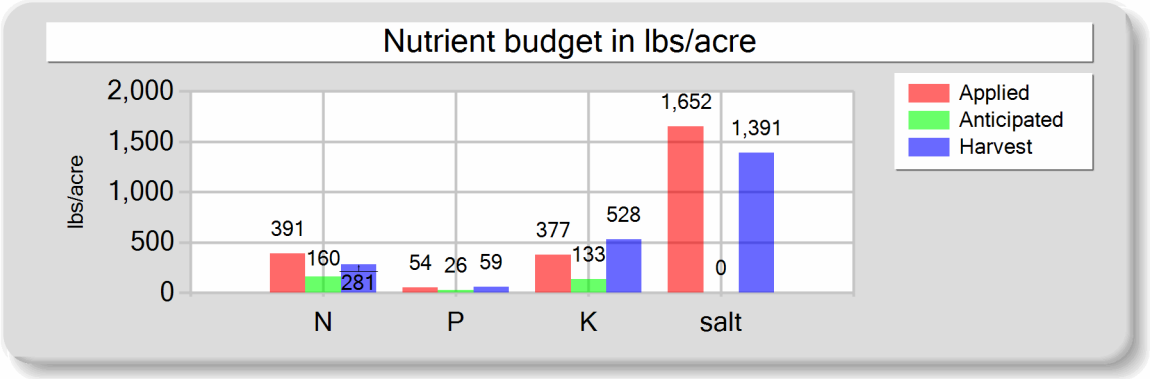
Fresh water applied	
26,056,878.00	*gallons*
959.59	*acre-inches*
39.98	*inches/acre*

Process wastewater applied	
1,481,142.00	*gallons*
54.55	*acre-inches*
2.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/10/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.05	22.79	69.50	0.00
Process wastewater	208.36	31.00	307.23	1,341.54
Fresh water	11.68	0.00	0.00	310.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	391.09	53.79	376.73	1,652.47
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	280.85	58.62	527.62	1,390.63
Nutrient balance	110.24	-4.84	-150.89	261.85
Applied to removed ratio	1.39	0.92	0.71	1.19

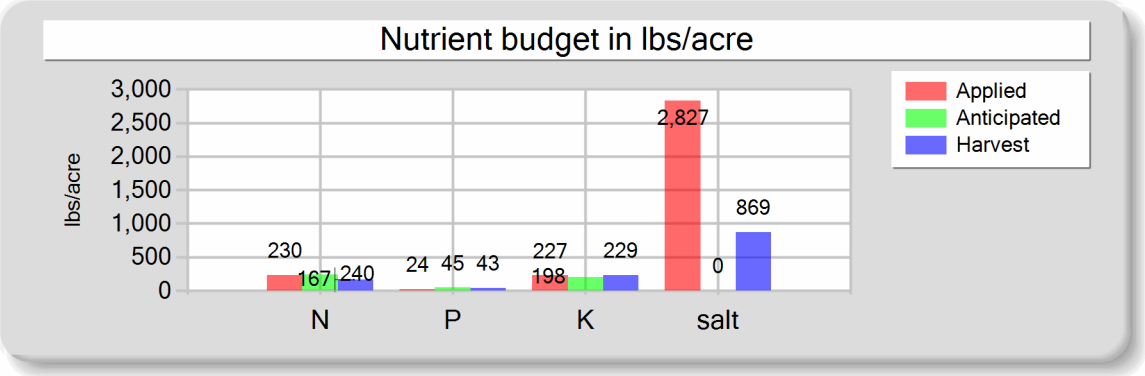
Fresh water applied	
8,309,225.00	*gallons*
306.00	*acre-inches*
8.05	*inches/acre*

Process wastewater applied	
1,001,571.00	*gallons*
36.88	*acre-inches*
0.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 05/13/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	146.93	23.97	226.76	799.33
Fresh water	76.15	0.00	0.00	2,027.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.08	23.97	226.76	2,826.86
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	166.99	42.78	229.09	869.44
Nutrient balance	63.09	-18.81	-2.33	1,957.42
Applied to removed ratio	1.38	0.56	0.99	3.25

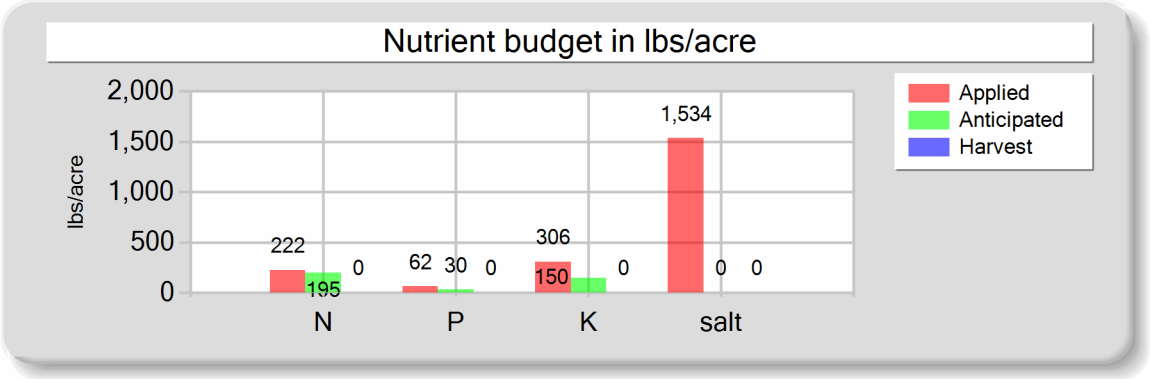
Fresh water applied	
54,182,062.00	*gallons*
1,995.34	*acre-inches*
52.51	*inches/acre*

Process wastewater applied	
635,143.00	*gallons*
23.39	*acre-inches*
0.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 01/23/2021: Almond, in shell

Field name: Field 9 Crop: Almond, in shell Plant date: 01/23/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	150.83	61.74	305.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	57.63	0.00	0.00	1,534.43
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	222.46	61.74	305.87	1,534.43
Anticipated crop nutrient removal	195.00	30.00	150.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	222.46	61.74	305.87	1,534.43
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied	
14,027,941.00	*gallons*
516.60	*acre-inches*
39.74	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Compost Manure

Sample and source description: 2022 Compost Manure

Sample date: 10/20/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 21.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	4,000.00	12,200.00							
DL	0.01	0.01	0.01							

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	8,800.00	43,600.00							
DL	0.01	0.01	0.01							

Corral Manure

Sample and source description: Corral Manure

Sample date: 11/14/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,000.00	10,700.00	29,100.00							
DL	0.01	0.01	0.01							

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 2022 - Dairy

Sample and source description: Lagoon Q4 2022 - Dairy

Sample date: 11/03/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	750.00	491.00	0.00	2.10	91.40	1,070.00								12,200.00	6,720
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4 2022 - Heifers

Sample and source description: Lagoon Q4 2022 - Heifers

Sample date: 11/03/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	181.00	77.00	0.00	1.00	59.90	581.00								4,060.00	2,520
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1 Dairy

Sample and source description: Lagoon Q1 Dairy

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	924.00	439.00	0.00	1.80	159.00	1,070.00	457.00	225.00	333.00	4,290.00	0.00	27.90	584.00	10,000.00	4,670
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

Lagoon Q1 Heifers

Sample and source description: Lagoon Q1 Heifers

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,140.00	488.00	0.00	3.10	214.00	2,670.00	576.00	259.00	505.00	6,160.00	0.00	41.40	1,750.00	16,000.00	10,100
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

Lagoon Q2 Dairy

Sample and source description: Lagoon Q2 Dairy

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	969.00	408.00	0.00	0.90	122.00	1,740.00								13,400.00	7,600
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2 Heifers

Sample and source description: Lagoon Q2 Heifers

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	544.00	165.00	0.00	0.90	135.00	1,020.00								6,590.00	5,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3 Dairy

Sample and source description: Lagoon Q3 Dairy

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,210.00	413.00	0.00	1.10	266.00	1,410.00								12,100.00	2,200
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4 Dairy

Sample and source description: Lagoon Q4 Dairy

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	238.00	148.00	0.00	0.00	80.20	820.00								8,830.00	7,050
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4 Heifers

Sample and source description: Lagoon Q4 Heifers

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	924.00	474.00	0.00	0.70	107.00	1,190.00								5,190.00	3,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

TID Canal

Sample description: TID Canal

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.40	0.00	6.40								284.00	
DL	0.01	0.01	0.01								10.00	

Domestic Well

DW Dairy

Sample description: DW Dairy

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	61.80	0.00	61.80								1,840.00	
DL	0.01	0.01	0.01								10.00	

Domestic Well

DW Heifers

Sample description: DW Heifers

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	57.20	0.00	57.20								1,750.00	
DL	0.01	0.01	0.01								10.00	

DW House

Sample description: DW House

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	69.80	0.00	69.80								1,640.00	
DL	0.01	0.01	0.01								10.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Espindula - 10/03/2022: Oats, silage-soft dough

Espindula Oats

Sample and source description: Espindula Oats

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	4,200.00	31,300.00		8.10
DL	0.01	0.01	0.01		1.00

Espindula - 05/13/2023: Corn, silage

Espindula Corn

Sample and source description: Espindula Corn

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,700.00	14,400.00		5.20
DL	0.01	0.01	0.01		1.00

Field 1 - 10/03/2022: Oats, silage-soft dough

Field 1 Oats

Sample and source description: Field 1 Oats

Sample date: 04/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	4,200.00	40,900.00		11.30
DL	0.01	0.01	0.01		1.00

Field 1 - 05/13/2023: Corn, silage

Field 1 Corn

Sample and source description: Field 1 Corn

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	3,100.00	16,600.00		6.30
DL	0.01	0.01	0.01		1.00

Field 11 - 10/03/2022: Oats, silage-soft dough

Field 11 Oats

Sample and source description: Field 11 Oats

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	4,200.00	39,200.00		10.50
DL	0.01	0.01	0.01		1.00

Field 11 - 05/13/2023: Corn, silage

Field 11 Corn

Sample and source description: Field 11 Corn

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,700.00	14,400.00		5.20
DL	0.01	0.01	0.01		1.00

Field 2 - 10/03/2022: Oats, silage-soft dough

Field 2 /Field 11 Oats

Sample and source description: Field 2 /Field 11 Oats

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	4,200.00	39,200.00		10.50
DL	0.01	0.01	0.01		1.00

Field 2 - 05/13/2023: Corn, silage

Field 2 Corn

Sample and source description: Field 2 Corn

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,700.00	14,400.00		5.20
DL	0.01	0.01	0.01		1.00

Field 3 - 11/10/2022: Oats, silage-soft dough

Field 3 Oats

Sample and source description: Field 3 Oats

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	4,100.00	39,100.00		10.30
DL	0.01	0.01	0.01		1.00

Field 3 - 05/13/2023: Corn, silage

Field 3 Corn

Sample and source description: Field 3 Corn

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,400.00	13,200.00		5.10
DL	0.01	0.01	0.01		1.00

Field 4 - 11/10/2022: Oats, silage-soft dough

Field 4 Oats

Sample and source description: Field 4 Oats

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	4,300.00	34,300.00		8.80
DL	0.01	0.01	0.01		1.00

Field 4 - 05/13/2023: Corn, silage

Field 4 Corn

Sample and source description: Field 4 Corn

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,900.00	13,600.00		5.50
DL	0.01	0.01	0.01		1.00

Field 5 - 11/10/2022: Oats, silage-soft dough

Field 5 Oats

Sample and source description: Field 5 Oats

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	4,300.00	34,300.00		8.80
DL	0.01	0.01	0.01		1.00

Field 5 - 05/13/2023: Corn, silage

Field 5 Corn

Sample and source description: Field 5 Corn

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,900.00	13,600.00		5.50
DL	0.01	0.01	0.01		1.00

Field 6 - 11/10/2022: Oats, silage-soft dough

Field 6 Oats

Sample and source description: Field 6 Oats

Sample date: 04/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,600.00	4,300.00	38,700.00		10.20
DL	0.01	0.01	0.01		1.00

Field 6 - 05/13/2023: Corn, silage

Field 6 Corn

Sample and source description: Field 6 Corn

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	3,100.00	16,600.00		6.30
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

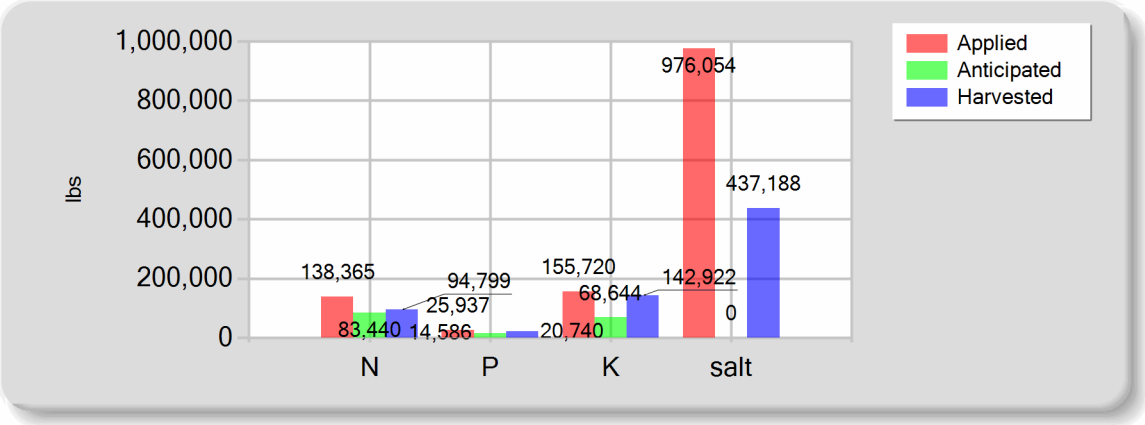
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

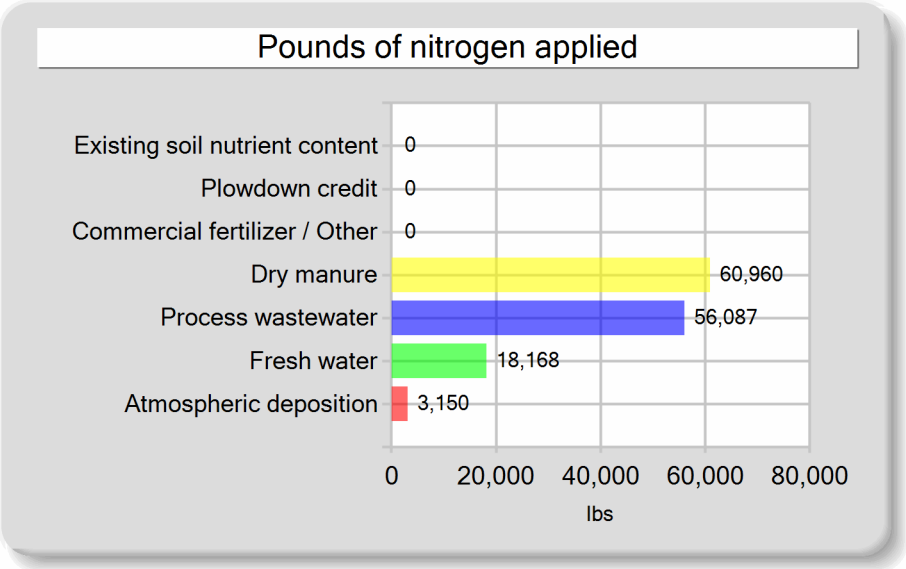
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

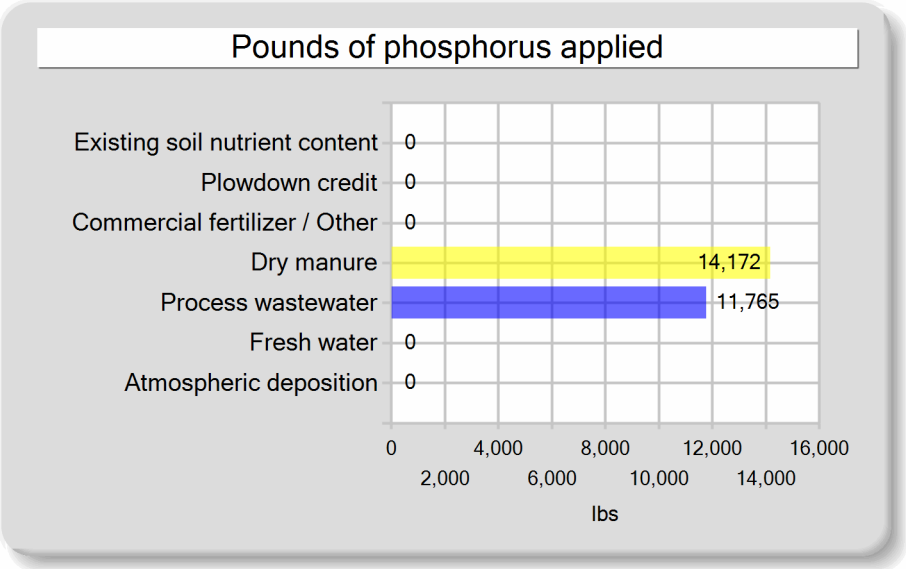
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	60,959.71	14,171.87	59,670.98	0.00
Process wastewater	56,087.13	11,765.41	96,049.30	492,328.54
Fresh water	18,168.09	0.00	0.00	483,725.50
Atmospheric deposition	3,150.00	0.00	0.00	0.00
Total nutrients applied	138,364.94	25,937.27	155,720.28	976,054.05
Anticipated crop nutrient removal	83,440.00	14,585.80	68,644.40	0.00
Actual crop nutrient removal	94,799.31	20,739.86	142,922.35	437,188.20
Nutrient balance	43,565.63	5,197.41	12,797.94	538,865.84
Applied to removed ratio	1.46	1.25	1.09	2.23

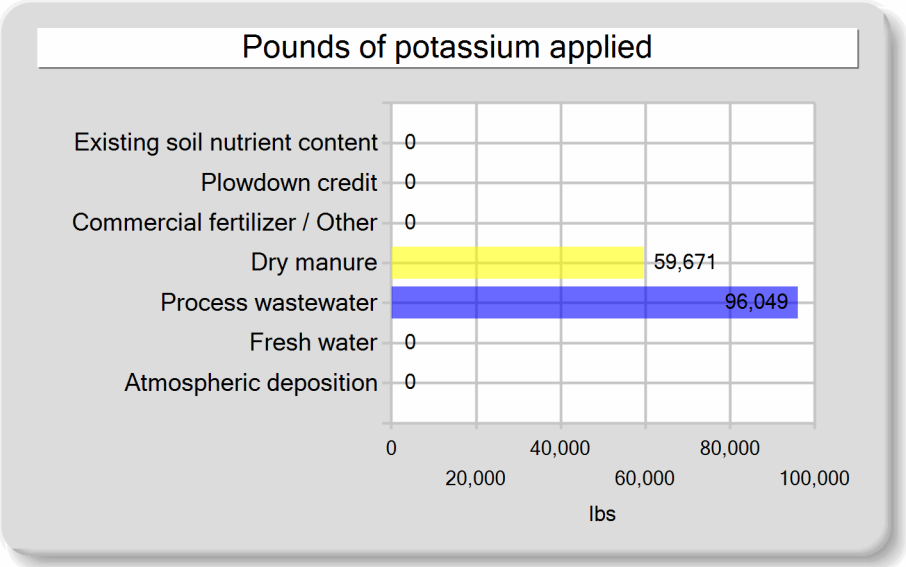
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

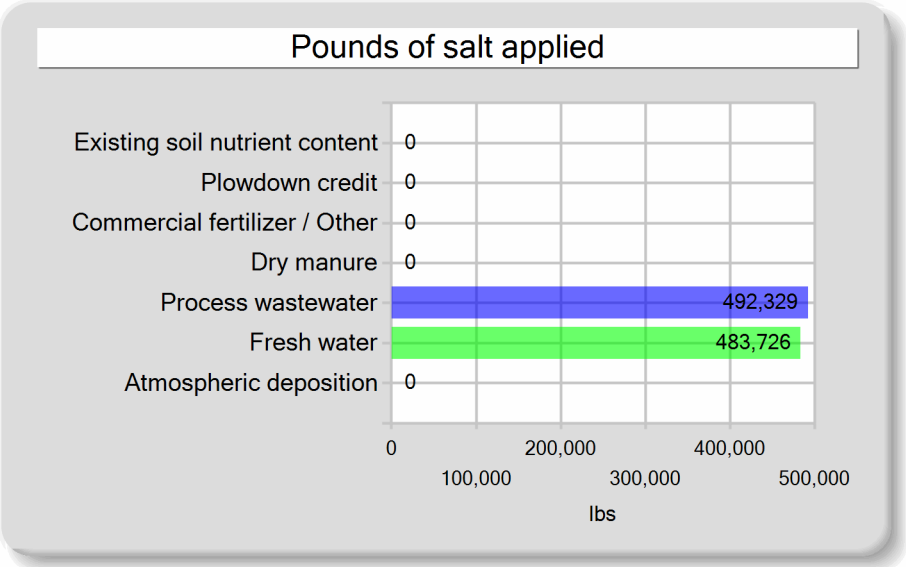


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Fields 9 and 10 are non producing almonds

ANNUAL REPORT VALIDATION INFORMATION

A. VALIDATION ERRORS

The following sections contain validation errors and should be reviewed before submitting the Annual Report :

1. Harvest Events

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuel M Silva	Adrian J Silva
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 2, 2024	Jul 2, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Adrian Silva

Name of Dairy Facility: Silvas Holsteins Dairy

Facility Address:

6706 Elaine RD	Turlock	Stanislaus	95380
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Adrian Silva (209) 595-1846
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Silva Holsteins Pipeline

Address of Hauling Company/Person:

6706 Elaine RD	Turlock	CA	95380
Number and Street	City	State	Zip Code

Contact Person: Adrian Silva (209) 595-1846
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Bill Ottman (209) 495-0534
Name Phone Number

5631 W Zeering RD	Ceres	CA	95307
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

5631 W Zeering RD	Ceres	95307
Address	City	Zip Code

Stanislaus
Street and nearest cross street (if no address) County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 06/04/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 2,880,000 gallons

Method used to determine volume of process wastewater:

gpm x sixty x hours ran

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

6/4/23

Operator Signature Date

6/4/23

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Adrian Silva

Name of Dairy Facility: Silvas Holsteins Dairy

Facility Address:

6706 Elaine RD	Turlock	Stanislaus	95380
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Adrian Silva (209) 595-1846
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Silva Holsteins Pipeline

Address of Hauling Company/Person:

6706 Elaine RD	Turlock	CA	95380
Number and Street	City	State	Zip Code

Contact Person: Adrian Silva (209) 595-1846
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Bill Ottman (209) 495-0534
Name Phone Number

5631 W Zeering RD	Ceres	CA	95307
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

5631 W Zeering RD	Ceres	95307
Address	City	Zip Code

Stanislaus
Street and nearest cross street (if no address) County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 06/15/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 2,520,000 gallons

Method used to determine volume of process wastewater:

gpm x sixty x hours ran

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

6/15/23

Operator Signature Date

6/15/23

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Adrian Silva

Name of Dairy Facility: Silvas Holsteins Dairy

Facility Address:

6706 Elaine RD	Turlock	Stanislaus	95380
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Adrian Silva (209) 595-1846
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Silva Holsteins Pipeline

Address of Hauling Company/Person:

6706 Elaine RD	Turlock	CA	95380
Number and Street	City	State	Zip Code

Contact Person: Adrian Silva (209) 595-1846
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Bill Ottman (209) 495-0534
Name Phone Number

5631 W Zeering RD	Ceres	CA	95307
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

5631 W Zeering RD	Ceres	95307
Address	City	Zip Code

Street and nearest cross street (if no address) Stanislaus
County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 07/22/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 2,592,000 gallons

Method used to determine volume of process wastewater:

gpm x sixty x hours ran

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

7/22/23

Operator Signature Date

7/22/23

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Adrian Silva

Name of Dairy Facility: Silvas Holsteins Dairy

Facility Address:

6706 Elaine RD	Turlock	Stanislaus	95380
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Adrian Silva (209) 595-1846
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Silva Holsteins Pipeline

Address of Hauling Company/Person:

6706 Elaine RD	Turlock	CA	95380
Number and Street	City	State	Zip Code

Contact Person: Adrian Silva (209) 595-1846
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Bill Ottman (209) 495-0534
Name Phone Number

5631 W Zeering RD	Ceres	CA	95307
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

5631 W Zeering RD	Ceres	95307
Address	City	Zip Code

Street and nearest cross street (if no address) Stanislaus
County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 10/15/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 2,160,000 gallons

Method used to determine volume of process wastewater:

gpm x sixty x hours ran

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

10/15/23

Operator Signature Date

10/15/23

Hauler Signature Date

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock CA 95380
00-0024405 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0753
Sampled By: M Dekeyrel
Requested By: Adrian Silva
Submitted Date: 12/12/23
Reported Date: 12/20/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW-Dairy	12/11/23	12:45	1840	1.84	61.8	ND		7.4
2	DW-House	12/11/23	12:55	1640	1.64	69.8	ND		7.7
3	DW-Heifers	12/11/23	13:06	1750	1.75	57.2	ND		7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Silva Holsteins Dairy
6706 Elaine Road
Turlock CA 95380
00-0024405 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0753
Sampled By: M Dekeyrel
Requested By: Adrian Silva
Submitted Date: 12/12/23
Reported Date: 12/20/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW-Dairy	12/11/23	12:45	**1840**	**1.84**	**61.8**	168.71	14.06
2	DW-House	12/11/23	12:55	**1640**	**1.64**	**69.8**	190.55	15.88
3	DW-Heifers	12/11/23	13:06	**1750**	**1.75**	**57.2**	156.16	13.01

Water for Crop Use	Total	Total	Nitrate
Nutrient Management	Salts	Salts	Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock, CA 95380

Account# 00-0024405
Account Manager: Ben Nydam
Submitted By: Adrian Silva

Received: 12/12/2023 15:55
Reported: 12/20/2023 15:47

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0753-01	DW-Dairy	Ag Water	M Dekeyrel		12/11/2023 12:45
23L0753-02	DW-House	Ag Water	M Dekeyrel		12/11/2023 12:55
23L0753-03	DW-Heifers	Ag Water	M Dekeyrel		12/11/2023 13:06

Default Cooler Temperature on Receipt °C: 0.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock, CA 95380

Account# 00-0024405
Account Manager: Ben Nydam
Submitted By: Adrian Silva

Received: 12/12/2023 15:55
Reported: 12/20/2023 15:47

Sample Results

Sample: DW-Dairy
23L0753-01 (Water)

Sampled: 12/11/2023 12:45
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.84**	mmhos/cm	0.01	1		12/14/23 15:22	SM 2510 B		BEL0614
Electrical Conductivity umhos	**1840**	umhos/cm	10.0	1		12/14/23 15:22	SM 2510 B		BEL0614
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 12:45	Field		BEL0503
Nitrate Nitrogen as NO3N	**61.8**	mg/L	0.1	1	10	12/13/23 08:08	EPA 300.0		BEL0349
Temperature	**25.0**	units	0.0	1		12/14/23 15:22	SM 4500-H+	H	BEL0614
pH	**7.4**	units	1.0	1		12/14/23 15:22	SM 4500-H+	H	BEL0614

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock, CA 95380

Account# 00-0024405
Account Manager: Ben Nydam
Submitted By: Adrian Silva

Received: 12/12/2023 15:55
Reported: 12/20/2023 15:47

Sample Results
(Continued)

Sample: DW-House
23L0753-02 (Water)

Sampled: 12/11/2023 12:55
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.64**	mmhos/cm	0.01	1		12/14/23 15:24	SM 2510 B		BEL0614
Electrical Conductivity umhos	**1640**	umhos/cm	10.0	1		12/14/23 15:24	SM 2510 B		BEL0614
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 12:55	Field		BEL0503
Nitrate Nitrogen as NO3N	**69.8**	mg/L	0.1	1	10	12/13/23 08:28	EPA 300.0		BEL0349
Temperature	**25.0**	units	0.0	1		12/14/23 15:24	SM 4500-H+	H	BEL0614
pH	**7.7**	units	1.0	1		12/14/23 15:24	SM 4500-H+	H	BEL0614

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock, CA 95380

Account# 00-0024405
Account Manager: Ben Nydam
Submitted By: Adrian Silva

Received: 12/12/2023 15:55
Reported: 12/20/2023 15:47

Sample Results
(Continued)

Sample: DW-Heifers
23L0753-03 (Water)

Sampled: 12/11/2023 13:06
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.75**	mmhos/cm	0.01	1		12/14/23 15:25	SM 2510 B		BEL0614
Electrical Conductivity umhos	**1750**	umhos/cm	10.0	1		12/14/23 15:25	SM 2510 B		BEL0614
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 13:06	Field		BEL0503
Nitrate Nitrogen as NO3N	**57.2**	mg/L	0.1	1	10	12/13/23 08:48	EPA 300.0		BEL0349
Temperature	**25.0**	units	0.0	1		12/14/23 15:25	SM 4500-H+	H	BEL0614
pH	**7.6**	units	1.0	1		12/14/23 15:25	SM 4500-H+	H	BEL0614

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock, CA 95380

Account# 00-0024405
Account Manager: Ben Nydam
Submitted By: Adrian Silva

Received: 12/12/2023 15:55
Reported: 12/20/2023 15:47

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0349										
Blank (BEL0349-BLK1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0349-BLK2)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0349-BLK3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0349-BLK4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0349-BS1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.7	90-110		
LCS (BEL0349-BS2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.5	90-110		
LCS (BEL0349-BS3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.1	90-110		
Duplicate (BEL0349-DUP1)		**Source: 23L0574-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L		4.6			1.92	10
Duplicate (BEL0349-DUP2)		**Source: 23L0733-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.6		0.1	mg/L		5.6			0.659	10
Duplicate (BEL0349-DUP3)		**Source: 23L0736-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	6.9		0.1	mg/L		6.8			0.930	10
Matrix Spike (BEL0349-MS1)		**Source: 23L0574-01**			Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	9.7		0.1	mg/L	5.000	4.6	102	90-110		
Matrix Spike (BEL0349-MS2)		**Source: 23L0733-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.9		0.1	mg/L	5.000	5.6	106	90-110		
Matrix Spike (BEL0349-MS3)		**Source: 23L0736-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	11.9		0.1	mg/L	5.000	6.8	102	90-110		
Reference (BEL0349-SRM1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		96.8	90-110		
Reference (BEL0349-SRM2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.1	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock, CA 95380

Account# 00-0024405
Account Manager: Ben Nydam
Submitted By: Adrian Silva

Received: 12/12/2023 15:55
Reported: 12/20/2023 15:47

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0349 (Continued)										
Reference (BEL0349-SRM3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.5	90-110		
Reference (BEL0349-SRM4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.8	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock, CA 95380

Account# 00-0024405
Account Manager: Ben Nydam
Submitted By: Adrian Silva

Received: 12/12/2023 15:55
Reported: 12/20/2023 15:47

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0614										
Blank (BEL0614-BLK1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.10		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	97.4		10.0	umhos/cm						
pH	3.2		1.0	units						
Blank (BEL0614-BLK2)					Prepared & Analyzed: 12/14/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	0.01		0.01	mmhos/cm						
Electrical Conductivity umhos	10.5		10.0	umhos/cm						
pH	4.4		1.0	units						
Blank (BEL0614-BLK3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Duplicate (BEL0614-DUP1)		**Source: 23L0743-03**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	2.12		0.01	mmhos/cm		2.12			0.165	10
pH	7.8		1.0	units		7.8			0.258	10
Electrical Conductivity umhos	2120		10.0	umhos/cm		2120			0.165	10
Duplicate (BEL0614-DUP2)		**Source: 23L0771-01**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.66		0.01	mmhos/cm		0.66			0.923	10
Electrical Conductivity umhos	664		10.0	umhos/cm		658			0.923	10
pH	7.5		1.0	units		7.6			0.529	10
Reference (BEL0614-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	458			umhos/cm	426.0		107	90-110		
Reference (BEL0614-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0614-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1150			umhos/cm	1000		115	90-110		
Electrical Conductivity umhos	1150			umhos/cm	1000		115	90-110		
Reference (BEL0614-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEL0614-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Silva Holsteins Dairy
6706 Elaine Road
Turlock, CA 95380

Account# 00-0024405
Account Manager: Ben Nydam
Submitted By: Adrian Silva

Received: 12/12/2023 15:55
Reported: 12/20/2023 15:47

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0614 (Continued)										
Reference (BEL0614-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEL0614-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0614-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0614-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

12/12/23 15:55 23L0753

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.
1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

Bill To: Acct No. 24405 Cons. 8

Purchase Order No. Results Needed By

Client: Silva Holsteins Dairy
Address: 6706 Elaine Road
City, State, Zip: Turlock CA 95380
Phone: Fax:
Cell/Email:

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Adrian Silva (209) 595-1846

Facility:

Date sampled: 12/11/23

Sampled by: mpekenrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: No. Bottles:

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test**)
(1) 1 L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-Dairy	Sampled From:	12/11/23	1245p	Ø	0.8
2. DW-House	Sampled From:	12/11/23	1250p	Ø	-0.3
3. DW-Heifers	Sampled From:	12/11/23	1060p	Ø	0.6
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	mpekenrel	CAS INC	12/11/23 10ep	12/12/23 @9:30am
Second	JCP	DLV	12-12-23 9:30	12-12-23 2:01
Third	MM	DLI	12/12 15:55	
Fourth				

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The pties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ ____ In
$ ____ Out

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	|	|	|							
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/12/23 15:55 23L0753

SEND OUT